Hah Min **Lew**

Ph.D. student · Korea University

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"Bridging AI research and scalable solutions for measurable real-world impact."

Biography

Ph.D. student at the Visual & General Intelligence (VGI) Lab, Korea University, supervised by Prof. Gyeong-Moon Park. Current research focus lies in Vision-Language-Action (VLA) models, Vision-Language-Navigation (VLN) models, and multimodal generative models.

Prior to academia, professional experience includes over three years as an AI Researcher at Klleon AI Research. Key contributions involved photo-realistic 3D human generation, enabling lifelike avatar movements and lip-sync. Holds both M.S. and B.E. degrees from DGIST, with a strong foundation in Deep Learning and Signal Processing, advised by Prof. Jae Youn Hwang.

Skills

Programming	**Python**, Bash, MATLAB, C, Java
Frameworks	**PyTorch**, TensorFlow, Keras, Scikit-learn, Pytorch3D
DevOps	**Docker**, Containerd, **Git**
Back-end Basics	Node.js
Front-end Basics	HTML, CSS, JavaScript
Languages	Korean, English

Education

Korea University — *Seoul, South Korea*

Ph.D. in Artificial Intelligence — *Mar. 2026 - present*

- Advisor: Prof. Gyeong-Moon Park

DGIST (Daegu Gyeongbuk Institute of Science and Technology) — *Daegu, South Korea*

M.S. in Electrical Engineering and Computer Science — *Mar. 2019 - Aug. 2022*

- Advisor: Prof. Jae Youn Hwang

DGIST (Daegu Gyeongbuk Institute of Science and Technology) — *Daegu, South Korea*

B.E. in School of Undergraduate Studies — *Mar. 2014 - Feb. 2019*

- Best Project Award

Experience

Klleon AI Research — *Seoul, South Korea*

AI Researcher — *Aug. 2022 - Feb. 2026*

- Accelerated inference using a training-free diffusion sampler (↑4.11×).
- Developed audio-driven 3DMM generation for virtual avatars with natural human-like expressions and movements. *(LVE ↓27.5%, FDD ↓28.9%, MEE ↓27.1%, CE ↓24.1%, Diversity ↑17.7%)*
- Developed a photorealistic head rendering model using Gaussian Splatting, outperforming 5 state-of-the-art models. *(MSE ↓59.96%, PSNR ↑4.41dB, SSIM ↑3.85%, LPIPS ↓38.16%)*
- Developed a Head Swap AI model with 212.7x speedup, reducing annual GPU costs by 99.53% (from $2.2M to $10.5K).
- Built large-scale multimodal data pipelines (9.41M+ frames from in-the-wild videos).
- Integrated external APIs (NVIDIA Omniverse Audio2Face, OpenAI ChatGPT, TTS) into a streaming avatar system, achieving 25-28 FPS performance.

Multimodal Biomedical Imaging and System Lab, DGIST — *Daegu, South Korea*

Graduate Researcher — *Mar. 2019 - Aug. 2022*

- Achievements: 6 SCIE publications, 7 international conferences, 9 projects, 4 patents, and 2 awards.
- Designed machine learning-based anomaly detection systems and generative models for biomedical imaging, focusing on multimodal, class-imbalance, and multi-task learning.
- Developed hardware-software integrated systems for application-specific use cases.
- Collaborative research with medical doctors from hospitals, including SNUH, SNUDH, and Yonsei Severance.

Publications

[C5] Personalized Audio-driven Whole-body Talking Avatars *Co-Author*

S. Lee, S. Moon, **Hah Min Lew**, J.-S. Kang, and G.-M. Park. CVPR 2026 *Jun. 2026*

[C4] Secondary Motion-Aware 3D Clothed Gaussian Avatars from Monocular Videos *Co-Author*

S. Lee, S. Moon, **Hah Min Lew**, J.-S. Kang, and G.-M. Park. ICLR 2026 *Apr. 2026*

[C3] GeoAvatar: Adaptive Geometrical Gaussian Splatting for 3D Head Avatar *First Author*

S. Moon*, **Hah Min Lew***, S. Lee, J.-S. Kang, and G.-M. Park. ICCV 2025 *Oct. 2025*

[C2] Towards High-fidelity Head Blending with Chroma Keying for Industrial Applications *First Author*

Hah Min Lew*, S.-M. Yoo*, H. Kang*, and G.-M. Park. WACV 2025 *Mar. 2025*

[C1] CSS-Net: Classification and Substitution for Segmentation of Rotator Cuff Tear *Co-Author*

K. Lee, **Hah Min Lew**, M. H. Lee, M. Kang, J. Kim, and J. Y. Hwang. ACCV 2022 *Dec. 2022*

[J6] Multi-task and Few-shot Learning-based Fully Automatic Deep Learning Platform for Mobile Diagnosis of Skin Diseases *Co-Author*

K. Lee, T. C. Cavalcanti, S. Kim, **Hah Min Lew**, D. H. Suh, D. H. Lee, and J. Y. Hwang. IEEE JBHI (IF: 7.021) *Jan. 2023*

[J5] Deep Learning-based Framework for Fast and Accurate Acoustic Hologram Generation *Co-Author*

M. H. Lee, **Hah Min Lew**, S. Youn, T. Kim, and J. Y. Hwang. IEEE TUFFC (IF: 3.267) *Dec. 2022*

[J4] Speckle Reduction via Deep Content-Aware Image Prior for Precise Breast Tumor Segmentation in an Ultrasound Image *Co-Author*

H. Lee, M. H. Lee, S. Youn, K. Lee, **Hah Min Lew**, and J. Y. Hwang. IEEE TUFFC (IF: 3.267) *Sep. 2022*

[J3] Intelligent Smartphone-based Multimode Imaging Otoscope for the Mobile Diagnosis of Otitis Media *Co-Author*

T. C. Cavalcanti, **Hah Min Lew**, K. Lee, S. Lee, M. K. Park, and J. Y. Hwang. Biomedical Optics Express (IF: 3.562) *Dec. 2021*

[J2] Ultrasonic Blood Flowmeter with a Novel Xero Algorithm for a Mechanical Circulatory Support System *First Author*

Hah Min Lew, H. Shin, M. H. Lee, S. Youn, H. C. Kim, and J. Y. Hwang. Ultrasonics (IF: 4.062) *Aug. 2021*

[J1] Forward-Looking Multimodal Endoscopic System Based on Optical Multispectral and High-Frequency Ultrasound Imaging Techniques for Tumor Detection *Co-Author*

J. Kim, **Hah Min Lew**, J. Kim, S. Youn, H. A. Faruque, A. N. Seo, S. Y. Park, J. H. Chang, E. Kim, and J. Y. Hwang. IEEE TMI (IF: 11.037) *Feb. 2021*

Projects

Audio-driven 3D Facial Animation for Realistic Facial Expressions and Motion *Seoul, South Korea*

Project Lead *Dec. 2024 - Feb. 2026*

- Developed a 3D facial animation framework for lifelike facial expressions and motion driven by audio inputs.
- Constructed a large-scale paired dataset of audio and 3DMM parameters (6.81M+ frames).
- Achieved superior performances compared to the SOTA method (LVE ↓27.5%, FDD ↓28.9%, MEE ↓27.1%, CE ↓24.1%, Diversity ↑17.7%).
- Accelerated diffusion sampling process 4.11x speed-up while preserving qualitative performances.
- Used skills: Python, PyTorch, Git.

Real-time Expressive 3D Chat Avatar System *Seoul, South Korea*

Project Lead *Apr. 2024 - Dec. 2024*

- Integrated NVIDIA Omniverse Audio2Face, OpenAI ChatGPT, and TTS APIs into a streaming avatar dialogue system with 25-28 FPS performance.
- Designed an emotion message queue protocol to enable natural emotional transitions and realistic facial expressions in avatars.
- Optimized Numpy-to-Tensor conversion and computations for live streaming, achieving a 13.5% speed improvement.
- Used skills: Python, PyTorch, Docker, Containerd, Git.

High-performance Real-time Head Swapping System *Seoul, South Korea*

Project Lead *Aug. 2022 - Apr. 2024*

- Led the development of a state-of-the-art head swapping framework, including data preprocessing pipelines, multi-GPU training, and efficient inference mechanisms.
- Built a high-quality dataset from 15,354 videos of 3,592 identities, processing 2.6M frames.
- Achieved a 212.7x inference speedup (from 10s/frame to 47ms/frame), reducing GPU resource requirements by 99.53%.
  - Reduced annual GPU costs from $2.2M+ (assuming 213 AWS EC2 g4dn.4xlarge instances) to approximately $10.5K (using a single instance).
- Achieved significant performance improvements over the SOTA method:
  - Metrics: PSNR ↑55.5%, LPIPS ↓91.8%, L1 ↓88.8%, SSIM ↑21.8%
  - Inference speed: 60.57 FPS (↑53.6%)
  - Computational efficiency: Parameters 8.92M (↓63.4%), MACs ↓33.0%
- Used skills: Python, PyTorch, Docker, Git, JavaScript, HTML, CSS.

Additional Projects

Image-to-Image Translation for High-resolution Gastrointestinal Imaging Project Lead — *Feb. 2021 - Sep. 2023*
Multitask Learning-based Network for Rotator Cuff Tear Segmentation Project Member — *Dec. 2021 - Dec. 2022*
Low-voltage CMUT-based Ultrasound Imaging for Medibots Project Member — *Sep. 2020 - Dec. 2022*
AI-powered Smartphone Imaging for Early Dental Caries Detection Project Lead — *Apr. 2020 - Feb. 2022*
ML-based Smartphone Imaging for Otitis Media Diagnosis Project Member — *Feb. 2020 - Jan. 2022*
2021 Laboratory-specialized Start-up Leader University Project Project Member — *Aug. 2021 - Jan. 2022*
Smart Monitoring System for Hip Implants Project Member — *Feb. 2019 - May. 2021*
Optimized Biomedical Monitoring System with a Time-efficient Algorithm Project Lead — *Mar. 2019 - Mar. 2021*
Technical Commercialization Activity Support for Bio Society Leadership Project Member — *May. 2020 - Dec. 2020*
Multimodal Data Registration and Analysis for Tumor Detection Project Member — *Mar. 2019 - Oct. 2020*
Ultrasonic Capsule Endoscopy Project Member — *Jun. 2019 - Jun. 2020*

Patents

COMPUTING DEVICE FOR HEAD SWAPPING — *Application*
Hah Min Lew, H. Kang, S.-M.Yoo, G.-M.Park (WO2025042068A1, KR1020230154188A) — *Feb. 2025*

MOBILE OTOSCOPE SYSTEM — *Patent*
J. Y. Hwang, T. C. Cavalcanti, **Hah Min Lew** (KR102648059B1) — *Mar. 2024*

ULTRASONIC BLOOD FLOW MEASURING APPARATUS AND METHOD THEREOF — *Patent*
J. Y. Hwang, **Hah Min Lew**, H. C. Kim (KR102514633B1) — *Mar. 2023*

BLADDER MONITORING APPARATUS AND METHOD FOR CONTROLLING BLADDER MONITORING APPARATUS — *Patent*
J. Y. Hwang, M. H. Lee, **Hah Min Lew** (KR102460829B1) — *Oct. 2022*

THREE-DIMENSIONAL DIAGNOSTIC SYSTEM — *Patent*
J. Y. Hwang, J. Kim, **Hah Min Lew** (KR102379481B1) — *Mar. 2022*

Awards & Scholarships

Awards

2021 **Outstanding Poster Award**, 2021 Student Conference at DGIST — *Daegu, South Korea*
2021 **Best Paper Award**, 2021 Spring Conference at KOSOMBE — *Remote, South Korea*
2017 **Best Project Award**, 2016 Undergraduate Group Research Project (UGRP) Program at DGIST — *Daegu, South Korea*

Scholarships

2014-2022 **Full Government Scholarships**, Full tuition exemptions and school expenses support — *Daegu, South Korea*

References

Gyeong-Moon Park — *Seoul, South Korea*
Associate Professor — *Mar. 2025 - present*
- Ph.D. in School of Electrical Engineering, KAIST, Seoul, South Korea. 2019.
- E-mail: gm-park@korea.ac.kr
- Office: Room #203B, Woo Jung Informatics Building

Jae Youn Hwang — *Daegu, South Korea*
Professor — *Sep. 2022 - present*
- Ph.D. in Biomedical Engineering, University of Southern California, Los Angeles, USA. 2009.
- E-mail: jyhwang@dgist.ac.kr
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